

Open Source IaaS Clouds: An Overview

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Abstract—Cloud computing enables on-demand provisioning of computing resources and services on a pay-as-you-go manner. Those resources can be provisioned rapidly with the help of virtualization. The central component that does the allocation of virtual machines on the cloud's physical resources is known as virtual machine scheduler. We have experienced the private cloud setup using the open source cloud technologies which can be scaled up with any number of compute nodes. Among many different offerings (service, platform, infrastructure), in this paper, we concentrate on infrastructure by presenting a comparative study of the open source cloud solutions. We have explored the components used for resource management and monitoring available in the open source cloud technologies such as Eucalyptus, OpenStack, and OpenNebula which have drawn the consideration of many people in the research community and the industry.

Keywords—Cloud; OpenStack; Eucalyptus; Virtual Machine Scheduling; Private Cloud

I. INTRODUCTION

As we can infer cloud computing is a continually unfolding archetype we couldn't just bind it down with a particular definition, the US National Institute of Standards and Technology's definition on Cloud Computing envelopes as the most vital feature to consider 'Cloud computing is a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction' [1]. Cloud consists of many data centers those encapsulates set of hosts. Host represents a physical computing node in a cloud. An organization can go with the public cloud if it wants to cut down the up-front investments on computing resources. If it has already invested the amount on computing resources, it can build a private cloud to improve the resource management and utilization.

Even though many services like Platform as a Service, Software as a Service are provided by the cloud, the base for all the services is Infrastructure as a Service. Infrastructure as a Service provides the virtualized computing resources to the users which can be adjusted according to the user's claim. Instead of purchasing the hardware outright, users can purchase the computing resources based on consumption like

electricity. How to efficiently allocate and manage the resources of the cloud is the ultimate goal of the Infrastructure as a Service. This paper discusses Private Cloud Setup using the open source software and the Virtual Machine Provisioning algorithms used in those technologies. The rest of the paper is organized as follows: Section 2 discusses the deployment models of cloud Section 3 deals with Private Cloud setup using the open source cloud technologies and virtual machine provisioning. Section 4 details the various virtual machine provisioning algorithms used in the open source cloud technologies. Section 5 summarizes the comparison of the open source cloud platforms and Section 6 gives the concluding remark of the whole paper.

II. DEPLOYMENT MODELS IN CLOUD

A. Public Cloud

As the name suggests public cloud is something that is probably out of premises managed by a third-party service provider, which is made available to the 'public' (i.e.) Anybody can use a public cloud be it a small to medium sized organization or a big organization. It emphasizes the pay-as-you-go model which implies that the enterprises have to pay only for those services that had been utilized by them, which is usually in the form of a monthly bill that will be generated by the service provider. Public cloud is envisioned to provide boundless computational resources via the internet to all enterprises. One of the key features of Public cloud is their moldable setting that lets their users configure their resources and also to choose what to hide and what not to hide from their clients. Most prominent instances of Public cloud are Amazon Elastic Compute Cloud (EC2) [2], IBM's Blue Cloud, Sun Cloud, GoogleAppEngine [3] and Windows Azure Services Platform [4].

B. Private Cloud

Private cloud is a model of cloud computing that evolves with a definite and shielded cloud setup where only the stated customer can benefit. It is a data center model, which is engineered in a way such that it can be inherited by a single enterprise. A private cloud provides elasticity, scalability, better computerization, supervision. The objective of a private cloud is to enjoy the offerings of the cloud architecture without giving up the security and control over the sensitive data. A

private cloud is similar to that of a public cloud, but it discards the exception over the control on the data. Private cloud is expensive when compared to the public version, which makes it unavailable for a lot of small and medium scale industries. They are more stringent when it comes to security and acquiescence, and keeping resources within the firewall. The resources of the private cloud are always drawn from distinct pool servers which can be hosted either internally or externally, but can be tapped from either a private connection or via an encrypted public connection, which strikes a contrast to the public cloud providers where all their resources lie in a shared pool of servers. Moving data into the Private cloud offers flexibility and convenience to people since they do not have to fear about third-party data management's trustworthy [5]. Creating these private clouds enable IT departments to focus on innovation for the business, reducing both capital and operational costs and automating the management of complex technologies [6].

C. Hybrid Cloud

Hybrid cloud is an infused concept of both Public and Private Cloud setup in order to carry out versatile tasks within the same enterprise. It is an environment where an organization handles few of its data in-house while outsourcing rest of its data via the public providers. For instance, an enterprise can use Amazons Elastic cloud to manage its archived data while it maintains its confidential data like customer information in-house with the help of a private cloud. This approach lets the consumers enjoy the benefits of scalability and cost-effectiveness from a public vendor without compromising on the critical information to the third-party vulnerabilities. This method proves to be cost-efficient than that of the private cloud set-up. Therefore, letting the organization achieve efficiency and cost minimization without forsaking security aspects. Another inevitable facet of Hybrid computing is the change involved should be minimized no matter how similar Public and Private clouds are matched there lies a remarkable difference in their underlying design which needs to be taken care of.

III. VIRTUAL MACHINE PROVISIONING AND THE PRIVATE CLOUD SETUP

Cloud framework changes the traditional computational infrastructure to virtual infrastructure. In the cloud, the services are not tied up with the physical resources. The advantages of virtualizations stated in [7] are it consolidates workloads to reduce hardware, power and space requirements. Virtual machine provisioning manages the virtual infrastructure with ease, offers these resources to the cloud users based their claim. This system is also known as thin provisioning, but the term is more relevant to physical computing scenarios than virtual environments. This is devised to consolidate resource management by letting a resource manager to process the incoming request for the various resources and provision them efficiently. The host allots resources to the virtual machines according to their

needs. Virtual Provisioning can intelligently cut down the cost incurred in power and cooling systems by putting the set of unproductive devices in a low power state. It is the sole responsibility of the provisioning manager to meticulously monitor each virtual machine inside the cloud environment to make sure that the quality of services are provided as required and there are no SLA violations. The provisioner also takes into account the efficiency of the devices while allotted inside a particular machine this efficiency can be calculated through various metrics. The sole purpose of the intermediary provisioner is to allocate resources to the virtual machines based on their request efficiently such that the load is balanced inside the data center and put the resources to their best use. While mapping a virtual machine to a physical machine, the assignment can be made according to the optimization criteria like compute-intensive, memory-intensive, affinity aware, power aware, load balanced, high performance, etc. [8]

For a research scholar, private cloud is a better option to experience the happening in a cloud environment. Because public clouds are all proprietary, even though we can avail the services but cannot able to know the workings in detail. A private cloud, on the other hand, allows the user to customize the algorithms, work out their problems. In our previous work [9] [10], we have implemented the private cloud setup using UEC and OpenStack. To deploy a minimal cloud infrastructure, at least, two dedicated systems are needed: a front end, one or more node(s) with minimum 2GB memory. In our Institution, we have set up the private cloud using a cloud controller with three compute nodes. The scalability of the cloud can be extended by adding any number of compute nodes with the cloud controller. Figure 1 represents the implementation diagram of OpenStack. Table 1 denotes the physical host's configuration used for setting up the private cloud using the open source cloud solutions.

Table.1 Physical hosts configuration

	CPU	Memory	Disk	OS
PH_1	Intel Xenon, 8 core, 64 bit X86, VT Enabled	16 GB	500 GB	Lubuntu 12.04
$PH_{2,3}$	Intel i7, 64 bit X86, VT Enabled	4 GB	250 GB	Lubuntu 12.04

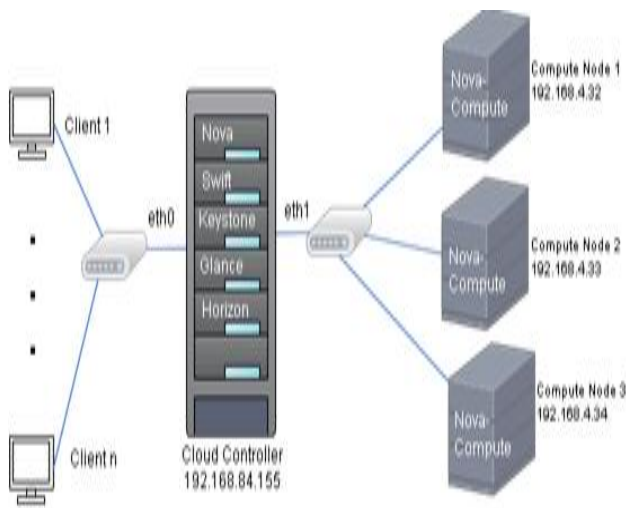


Fig. 1. Implementation diagram of cloud controller and compute node in OpenStack

IV. OPEN SOURCE CLOUD TECHNOLOGIES AND THEIR VIRTUAL MACHINE PROVISIONERS

A. OpenNebula

OpenNebula is a free and open-source virtual machine provisioner, which is highly bendable and notably feature-rich [11]. It manages the resources available in the resource pool and also takes care of the provisioning of the various resources to the Virtual machines efficiently. It supports public, private, hybrid implementations of the cloud. It acts as a scheduler which provisions the resources from the resource pool to the virtual machines. It provides assorted schemes when it comes to provisioning a virtual machine with resources. OpenNebula has an in-built provisioner, which allows the virtual networks to be corresponded to the physical ones. By which the physical network is broken down into smaller chunks of the network where the virtual machines can be allotted. In this way, we can ensure that communication between the virtual machines inside the network.

As mentioned earlier OpenNebula is a free and open-source resource management toolkit, which is used to build Infrastructure as a Service (IaaS), by provisioning virtual machines by separating the resource pool into clusters and data centers. Cloud APIs like REST, OGF OCCl, AWS EC2 API comes inherited within OpenNebula and it also holds into numerous hypervisors, Xen, KVM, VMware etc., The Architecture of OpenNebula is a cell-based structure wherein we can integrate our scheduling algorithms, its frontier lies an intermediary physical machine which connects the chunks of clusters which in turn contains a set of virtual machines. The default scheduler inside the OpenNebula provides a rank based scheduling scheme where the virtual machines are allotted resources based on ranks, however as the OpenNebula platform is flexible enough to accommodate the user's

scheduling schemes inside the clusters, the users can modify the scheduling schemes to suit their needs.

OpenNebula scheduling component is designed in a generic way, so it is highly modifiable and can be easily replaced by third-party developments. OpenNebula comes with a match-making scheduler. The main motive behind this Rank Based Scheduling Policy is to prioritize the resources which are more suited for the virtual machine. From the set of available hosts, the scheduler filtered out all the hosts those do not have enough resources to accommodate the virtual machines. Then on the set, it applies the RANK expression to evaluate the resources most suitable for the virtual machine. Resources with higher rank are used to allocate the virtual machines.

B. OpenStack

OpenStack is a cloud operating system, a new management layer that adds automation and control for a pool of resources such as computation, storage, and network, empower administrators and users via self-service portals and empower developers to make apps cloud-aware via APIs [12]. OpenStack is a cloud framework or a platform, which is, in turn, a collection of different components. These components usually are used to deploy as an Infrastructure as a Service solution where it provides a platform to deploy public and private cloud. OpenStack is an open-source and free software which contains various components which support different users for their varied needs. Open stack allows multiple instances of virtual machines which can handle the varied process on the fly enabling to scale-up tremendously by allowing many processes to carry out concurrently. This also leverages the cloud environment by supporting it in a way such that it can handle few or more users concurrently. Taking advantages of the fact that the open stack is open-source and free thousands of developers all over the world could collaborate and work on improving the framework thus by ensuring that the framework is the most secure, strong and contribute to a large community. OpenStack is a composite of various components such as Nova which handles the various threads of virtual machines and is the major computing engine, Swift, which is responsible for the storage of objects and other files, Glance which provides image services i.e., virtual page services to the virtual machine instances, Horizon is a graphical user interface component that is the dashboard component of the OpenStack. Figure 2 describes the cloud environment made by the OpenStack.

The schedulers available in OpenStack are FilterScheduler, ChanceScheduler, SimpleScheduler and MultiScheduler. SimpleScheduler finds the least loaded host. ChanceScheduler randomly selects from the list of filtered hosts. Another option MultiScheduler is there which apply different schedulers for nova-compute and nova-volume. Inside OpenStack, the default scheduler is the filter scheduler for provisioning virtual machines. Filter scheduler usually maintains a dictionary of

unfiltered hosts for which the various filters will be applied and virtual machines are assigned. Filters can also be compared with weighting; in the former the algorithms used can be quite flexible. The various filter algorithms in practice are ComputeFilter-which filter the hosts according to the required compute units, RamFilter-filter the hosts according to the ram required by the VM, CoreFilter- filters the host according to the number of cores required etc..

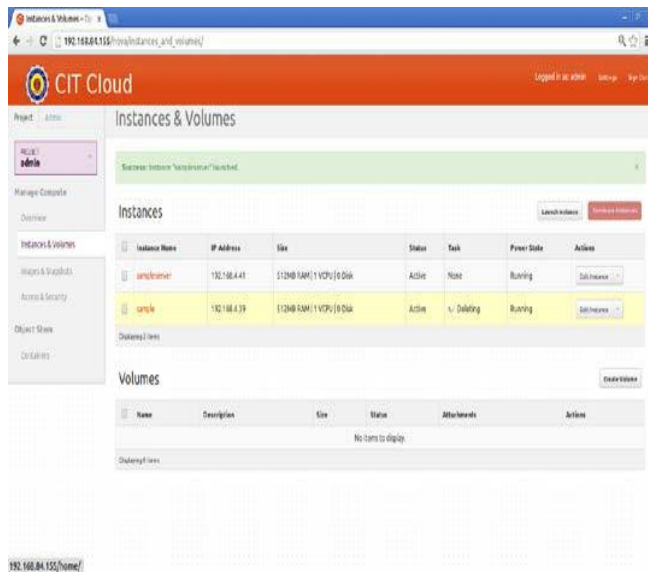


Fig. 2. Virtual Machines available in the OpenStack Cloud Environment

C. Ubuntu Enterprise Cloud

Eucalyptus is an open-source and free Infrastructure as a Service private cloud framework for building private and hybrid cloud computing environments. Eucalyptus stands for Elastic Utility Computing Architecture for channeling the tasks to the resource pool [13]. It enables to scale up and down the various resources in the resource pool according to the utility changes dynamically. It contains five major components within them they are (1) The Cluster Controller (CC) gathers all the information about the different schedules inside the specific node (2) The Storage Controller (SC) controls the retrieving and posting of storage information of the virtual machine images and user data. (3) The Elastic Block Storage (EBS) (4) Cloud Controller (CC) it directs the request from the users and the administrators towards the Cluster Controller it makes the high-level decisions on the scheduling and finally the (5) The Node Controller (NC) this controls execution and termination of the virtual machine instances on the host machines. In Figure 3 UEC image store is shown.

Scheduling in Eucalyptus determines the process by which Virtual Machines are allocated to the nodes. This is done to balance the load on the resource network consisting of numerous host machines. In Eucalyptus throughput, fairness in waiting time and response time are considered important factors. By throughput we mean at the give time number of VM's that can be successfully allotted inside various host,

Fairness in Waiting time denotes the time period that a VM has to wait in order for it to be scheduled with resources, Response time is the amount of time taken by the provisioner to provision the VM that has requested for the resources. The Scheduling Algorithm that is most popularly used in the cloud computing arena is the Greedy Algorithm which is a simple algorithm that tries to fit the requesting VM with the host machine that it first finds in the network. As we can infer Greedy Algorithm is one of the simplest algorithms when it comes to scheduling but, on the flip-side, it also has a very major drawback called underutilization of the resources in the resource pool. Suppose consider three host machines A,B,C in the resource pool each having the capacity of 100%,40%,70% while an incoming request is placed for 40% according to the greedy algorithm the scheduler will try to fit in the request inside the first node that it sees which, is host machine A. As far as efficiency is considered it is not a best practice and intuitively we can say that the host machine B best suits the need. But hence, the greedy approach is being considered inside Eucalyptus this is how the resources are being scheduled.

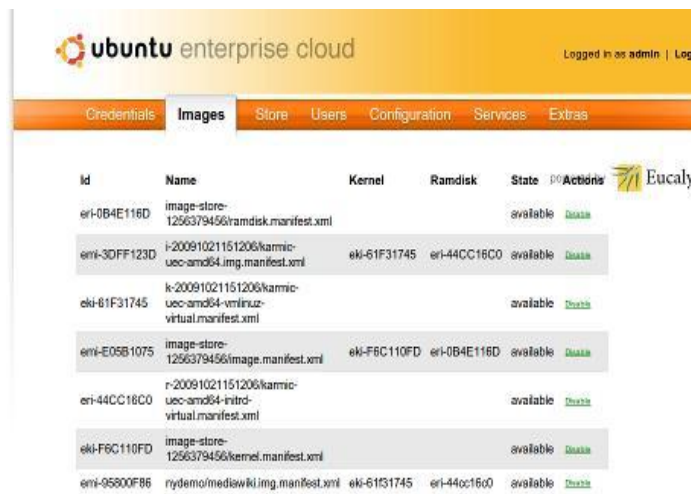


Fig. 3. Snapshot showing Image Store of UEC

V. COMPARISON OF OPEN SOURCE CLOUD PLATFORMS

Based on our study Table 2 summarizes the comparison analysis of cloud solution platforms Eucalyptus, OpenStack and OpenNebula considering their general aspects. In our previous work, we have developed a tool in OpenStack cloud environment which is helpful for the cloud administrator. It alerts the administrator about the resource utilization of each host [14].

Table 2. Comparison of cloud solution platforms

	Eucalyptus	OpenNebula	OpenStack
Installation Effort	Difficult	Medium	Difficult
Open Source License	Linux Open Source	Apache 2.0	Apache 2.0
Administration	Web UI, CLI	Web UI, CLI	Web UI, CLI
Language	Java, C, Python, Perl, Shell	C, Ruby, Shell Scripts, Java	Python, Shell Scripts
Hypervisor Support	Xen, KVM	Xen, KVM	KVM, XenServer, Hyper V, VMWare (Focus on KVM)
Cloud Interface	Amazon EC2 API, Proprietary API, OCCl API	OCCI API, Amazon EC2 API, LibVirt API, Proprietary API	Amazon EC2 API, Proprietary API, OCCl API
EC2 Compatibility	Yes	Yes	Yes (Nova EC2 API)
Virtual Network	Static and Dynamic	Static and Dynamic	Static and Dynamic
VM placement policies	Greedy and round robin	Match Making - Initial placement based on rank policy	Filter Scheduler
VM Placement Configuration Policies	No	Can customize according to the user's need	Can customize according to the user's need
Live Migration	No	Yes	Yes
Load Balancing	Yes	Yes	Yes

VI. CONCLUSION AND FUTURE SCOPE

The basic requirement for setting up a private cloud environment is not at all expensive. Cloud computing has the potential to revolutionize the computing facilities thus institutions can use it in a cost effective way. There are different open source technologies which are available to build private cloud within campuses. In this paper, we have done this comparison from the user's standpoint, the needs they are

expecting while choosing an IaaS cloud technologies. In future, we are going to develop a virtual machine scheduling algorithm to schedule the virtual machines in the host to make the hosts better resource utilization.

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